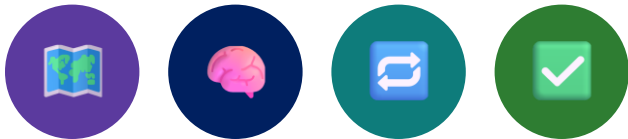


MODULE 45

Infrastructure as Code with AI Assistance

Provision infrastructure with Terraform, configure it with Ansible, and use AI assistance to accelerate, validate, and improve every decision -- with humans always accountable.







MODULE 45 OVERVIEW

Provision infrastructure with Terraform, configure it with Ansible, and use AI assistance responsibly -- with humans accountable for every apply.

The Northstar CRM team wants faster, repeatable environment setup for dev/test/stage -- but AI-generated infrastructure can be syntactically plausible while insecure, destructive, expensive, or non-idempotent.

DURATION & LAB



1 Hour Theory

Infrastructure as Code, Terraform provisioning, Ansible configuration management, and AI-assisted review discipline.



2 Hours Lab

lab45-crm: AI-drafted Terraform + Ansible sketches for Northstar CRM, validated, corrected, and documented.



Scenario

Northstar CRM non-production environments (crm-dev / crm-test) -- never real customer data in IaC.

MODULE ARC



Understand IaC

Why automation matters, what IaC is, and the end-to-end workflow.



Master Terraform & Ansible

Provisioning, state, configuration management, and idempotence.



Use AI Responsibly

Draft with AI, review critically, and document every decision in the lab.

KEY TAKEAWAY Total time: 1 hour theory + 2 hours hands-on lab = 3 hours. Syntactically valid, AI-generated Terraform that opens a public database still fails the lab -- humans stay accountable.

WHY INFRASTRUCTURE AUTOMATION MATTERS

Modern applications demand infrastructure that is consistent, scalable, secure, and available on demand -- manual processes can't keep up.

The Challenge with Manual Infrastructure	The Solution: Infrastructure Automation
Time-consuming and error-prone.	Automate provisioning and configuration.
Inconsistent environments.	Consistent, repeatable environments.
Difficult to scale quickly.	Scale infrastructure in minutes.
Hard to track changes and audit.	Version-controlled and auditable changes.
Higher risk of security misconfigurations.	Built-in security and compliance.
Increased operational costs.	Lower costs, higher efficiency.

BUSINESS IMPACT

**Faster Time to Market**

Teams focus on features, not managing servers.

**Scalability on Demand**

Provision and scale to meet business needs.

**Reliability & Security**

Consistent configs reduce error and risk.

**Cost Optimization**

Right-size infrastructure to reduce waste.

KEY TAKEAWAY With AI assistance, you can write better IaC code, detect misconfigurations, and automate with confidence. Automate today, deliver value every day.

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to provision, configure, and manage infrastructure using Terraform, Ansible, and AI assistance.

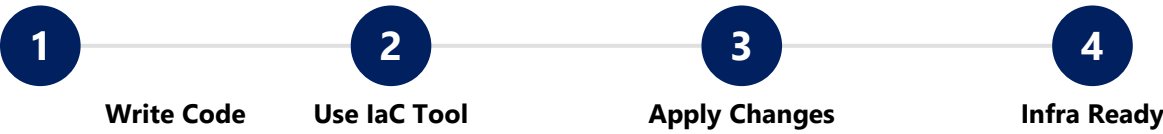
- **1. Understand IaC** — explain the concept, benefits, and workflow of Infrastructure as Code and why automation matters.
- **2. Work with Terraform Fundamentals** — understand Terraform architecture, providers, resources, and configuration files.
- **3. Execute the Terraform Workflow** — use terraform init, plan, apply, and destroy to provision and manage infrastructure.
- **4. Manage Terraform State** — understand state files, remote state concepts, and best practices for state management.
- **5. Understand Ansible Fundamentals** — explain Ansible architecture, inventory, playbooks, and how Ansible works.
- **6. Automate Configuration Management** — create Ansible playbooks to automate configuration tasks with idempotent operations.
- **7. Combine Terraform and Ansible** — integrate Terraform for provisioning and Ansible for configuration into one workflow.
- **8. Apply Best Practices** — implement industry best practices for secure, scalable, repeatable infrastructure automation.

KEY TAKEAWAY You will gain practical skills to automate infrastructure, improve reliability, reduce manual effort, and accelerate delivery with AI-powered assistance.

WHAT IS INFRASTRUCTURE AS CODE?

IaC is the practice of managing and provisioning infrastructure through machine-readable code instead of manual processes.

HOW IaC WORKS



Define infra in code -> the tool reads it -> provisions or updates infra -> deployed, consistent, and repeatable.

WHY IT MATTERS

- **Consistency** — same code = same environment every time.
- **Automation** — eliminates manual steps and human errors.
- **Scalability** — provision and scale resources quickly and reliably.
- **Version Controlled** — track changes, review, and roll back using Git.
- **Cost Efficiency** — optimize resource usage and reduce operational costs.

MANUAL VS INFRASTRUCTURE AS CODE

Manual Infrastructure	Infrastructure as Code
Manual, repetitive tasks	Automated and repeatable
Prone to human error	Reduced errors
Inconsistent environments	Consistent environments
Hard to track changes	Version controlled
Slow to scale	Fast and scalable

EXAMPLES OF IaC TOOLS

Terraform

Ansible

AWS CloudFormation

Pulumi

Azure Bicep

Key idea: you describe infrastructure in code (declarative or imperative); the tool reads it and automatically creates, updates, or destroys the infrastructure for you.

KEY TAKEAWAY IaC brings software engineering best practices -- code, version control, automation, testing, deployment -- to infrastructure. Write once, use everywhere.

BENEFITS OF INFRASTRUCTURE AUTOMATION

Infrastructure automation enables teams to deliver consistent, secure, and scalable environments faster while reducing costs and manual effort.



1. Speed & Agility

Provision and configure infrastructure in minutes, not hours or days.



2. Consistency

Use code to ensure the same environment every time across dev/test/prod.



3. Reliability

Reduce human error and misconfigurations with repeatable, versioned infra.



4. Scalability

Easily scale resources up or down without manual intervention.



5. Security & Compliance

Enforce policies as code; detect drift and remediate automatically.



6. Cost Efficiency

Optimize resource usage, eliminate waste, only pay for what you need.



7. Visibility & Auditability

Track every change and audit infrastructure like application code.



8. Better Collaboration

Enables collaboration between operations, developers, and security teams.

KEY TAKEAWAY Automate once, use everywhere, deliver value always. Less manual work, more innovation, better outcomes -- forming the backbone of modern DevOps and cloud-native environments.

IaC WORKFLOW OVERVIEW

Infrastructure as Code follows a consistent workflow from writing code to managing and maintaining your infrastructure.



A continuous feedback loop: after Monitor & Maintain, the cycle returns to Write Code -- not a one-time event.

KEY PRINCIPLES

- **Declarative** — define the desired state, not the steps.
- **Idempotent** — apply changes repeatedly with the same result.
- **Versioned** — track all changes like application code.
- **Reproducible** — recreate environments consistently.
- **Automated** — reduce manual work and human errors.

BUSINESS VALUE

- Faster delivery.
- Consistent & reliable environments.
- Lower costs.
- Better visibility and stronger collaboration.
- Audit & compliance built in.

Outcome: infrastructure that is consistent, scalable, secure, and delivered efficiently with code.

KEY TAKEAWAY IaC brings DevOps best practices to infrastructure: Code, Automate, Validate, Deploy, Operate, Improve -- a continuous feedback loop, not a one-time event.

AI ASSISTANCE IN INFRASTRUCTURE AS CODE

AI accelerates drafting Terraform and Ansible -- but humans remain accountable for exposure, cost, state security, and idempotence.

WHERE AI ASSISTANCE HELPS

- **Code Generation** — scaffold IaC templates, modules, and boilerplate.
- **Code Explanation** — explain existing IaC code and resources.
- **Refactoring & Optimization** — improve structure, reduce cost and complexity.
- **Policy & Compliance** — generate and validate policies, checks, and guardrails.
- **Troubleshooting** — analyze errors, suggest fixes and best practices.
- **Documentation** — auto-generate documentation and diagrams.

THE HUMAN ACCOUNTABILITY GATE

- Write bounded, contract-first prompts that state assumptions and forbid secrets/public exposure.
- Review generated Terraform/Ansible critically -- 'it linted' is not a review.
- Never apply because 'AI said so' -- a human reads every plan action.
- Reject or harden at least one unsafe AI suggestion, with rationale.
- Document every AI decision in a dated AI review record.

Example: a bounded, contract-first prompt

```
Generate Terraform for a non-prod Kubernetes namespace crm-${environment}
with labels application=crm. No public LoadBalancer/DB. No plaintext secrets.
Pin provider versions. List every assumption made.
Include a human review checklist alongside the generated code.
```

KEY TAKEAWAY Syntactically valid Terraform that opens a public database still fails review. AI drafts scaffolding; humans stay accountable for every apply.

TRY IT YOURSELF — EXERCISE 1: INFRA CONTRACT

Reinforces: writing a bounded infra contract before any AI prompt -- what's allowed, what's forbidden, before Terraform begins.

STEPS (notes/lab45-infra-contract.md)

Step	What To Do
1 — Contract Fields	State env names (crm-dev/crm-test), region, network, runtime, DB posture, tags, cost limits.
2 — Forbidden List	Write the reference table: allowed (sketches, .tfvars.example) vs forbidden (real keys, tfstate, PII).
3 — Tags	Propose tags: application=crm, environment=dev, owner= <your note>.
4 — Data Rule	State explicitly: fixtures CUS-1001 / CUS-1002 stay in app labs -- never in Terraform/Ansible.

Reference: allowed vs forbidden in IaC

```
Allowed:  network/runtime sketches, tfvars.example, inventory.example.yml, tags
Forbidden: real cloud keys, terraform.tfstate, customer PII, unreviewed public DB
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 1 before continuing: exercises/exercise-01-infra-contract.md (workspace: examples/module-45-exercises).



WHAT IS TERRAFORM?

Terraform is an open-source Infrastructure as Code tool by HashiCorp that lets you define, provision, and manage infrastructure using declarative configuration files.

KEY IDEA

- You describe the desired state of your infrastructure in code, and Terraform figures out how to create or update it.

Example resource block

```
resource "aws_instance" "web" {  
  ami           = "ami-0c55b159cbfafa1f0"  
  instance_type = "t3.micro"  
  tags = {  
    Name = "web-server"  
  }  
}
```

WORKS WITH 100+ PROVIDERS

AWS

Azure

Google Cloud

Kubernetes

VMware

...and many more

WHY USE TERRAFORM?



Multi-Cloud

Works with AWS, Azure, Google Cloud, and more.



Consistent

Repeatable deployments every time.



Version Controlled

Store infra as code in Git for collaboration.



Cost Efficient

Automate and optimize to reduce manual effort.

KEY TAKEAWAY Terraform helps teams build, change, and version infrastructure safely and efficiently using code -- enabling automation, consistency, and reliability at scale.

TERRAFORM ARCHITECTURE

Terraform uses a modular architecture to define, provision, and manage infrastructure across multiple cloud providers.



TERRAFORM CORE RESPONSIBILITIES

- **Parses HCL** — reads and validates .tf files written in HashiCorp Configuration Language.
- **Creates Execution Plan** — shows what will be created, changed, or destroyed before applying.
- **Builds Dependency Graph** — determines resource relationships and correct order of operations.
- **Manages Resource Lifecycle** — creates, updates, and deletes resources to reach the desired state.
- **Tracks Infrastructure State** — stores and refreshes state to keep infrastructure in sync.

COMPONENTS

Component	Role
Terraform CLI	Command-line interface for running Terraform commands.
Terraform Core	Dependency graph, planning, resource management.
Terraform State	Stores current state; maps real resources; enables operations.
Provider Plugins	Translate HCL into cloud-specific API calls.

KEY TAKEAWAY Terraform separates infrastructure logic from cloud-specific implementations using provider plugins, enabling consistent, multi-cloud automation.

PROVIDERS AND RESOURCES

Terraform uses providers to interact with APIs of different platforms. Providers expose resources that represent infrastructure objects.

KEY CONCEPTS

- **Provider = Plugin** — communicates with a platform (cloud, SaaS, or on-prem).
- **Resource = Infra Object** — the individual objects Terraform manages (VM, VPC, bucket, DB).
- **Multiple Providers** — use several providers in one configuration for multi-cloud.
- **Lifecycle Management** — Terraform handles create, read, update, delete, and keeps state in sync.

EXAMPLE RESOURCES BY PROVIDER

Provider	Resources (examples)
AWS	EC2, VPC, S3, RDS, IAM
Azure	Virtual Machines, Storage Account, Virtual Network
Google Cloud	Compute Engine, Cloud Storage, GKE
Kubernetes	Pods, Deployments, Services, ConfigMaps

HOW TERRAFORM WORKS (HIGH-LEVEL FLOW)



Providers translate Terraform configuration into platform-specific API calls, while resources represent the infrastructure objects being created and managed.

GITHUB PROVIDER (BEYOND CLOUD)

Providers aren't only clouds: 1000+ providers exist for databases, monitoring, networking, DevOps tools, and SaaS platforms like GitHub (repositories, teams, actions).

KEY TAKEAWAY Providers translate Terraform configuration into platform-specific API calls, while resources are the infrastructure objects being created and managed.

TERRAFORM CONFIGURATION FILES

Terraform uses Human-Readable Configuration Language (HCL) in .tf files to define infrastructure in a declarative, version-controlled way.

ANATOMY OF A .tf FILE

- **Provider Block** — specifies the cloud or platform to use.
- **Resource Block** — defines infrastructure resources to create or manage.
- **Variable** — makes values reusable and configurable.
- **Output** — exposes information about provisioned resources.

example.tf

```
terraform {
  required_version = ">= 1.6"
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
}

resource "aws_s3_bucket" "data" {
  bucket = var.bucket_name
  tags = {
    environment = var.environment
  }
}

output "bucket_name" {
  value = aws_s3_bucket.data.bucket
}
```

COMMON FILE STRUCTURE

File	Purpose
main.tf	Main resources
providers.tf	Provider config
variables.tf	Input variables
outputs.tf	Output values
.tfvars	Variable values

BEST PRACTICES

- Use meaningful resource names and tags.
- Use variables for values that change per environment.
- Organize code using modules and folders.
- Never hardcode secrets in .tf files.
- Always use Git for version control and reviews.

KEY TAKEAWAY Terraform configuration files are the source of truth for your infrastructure. Write once, use everywhere -- HCL is declarative and version-controlled.

TERRAFORM INIT

The terraform init command initializes a Terraform working directory -- always the first command to run, and safe to run multiple times.



1. Initializes

Creates the .terraform directory to store plugins and modules.



2. Downloads Plugins

Downloads required provider plugins defined in the configuration.



3. Installs Modules

Downloads and installs modules from the Terraform Registry or other sources.



4. Configures Backend

Prepares the backend for storing the state file (local or remote).

\$ terraform init

```
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.30.0...
```

```
Terraform has been successfully initialized!
```

WHEN TO USE

- When starting a new Terraform project.
- After cloning a repository.
- When provider or module requirements change.
- When backend configuration changes.

KEY TAKEAWAY terraform init is always the first command to run and sets up everything needed to work with Terraform. It is safe to run multiple times.

TERRAFORM PLAN

The terraform plan command creates an execution plan -- a read-only preview of what Terraform will do before making any changes.



1. Reads Config & State

Loads .tf files and the current state from the backend.



2. Compares

Compares the desired state (config) with the current state.



3. Creates Execution Plan

Determines the actions needed to reach the desired state.



4. Safe & Non-Destructive

No changes are made during the plan operation.

\$ terraform plan

```
# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami           = "ami-0c55b159cbfafe1f0"
  + instance_type = "t3.micro"
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

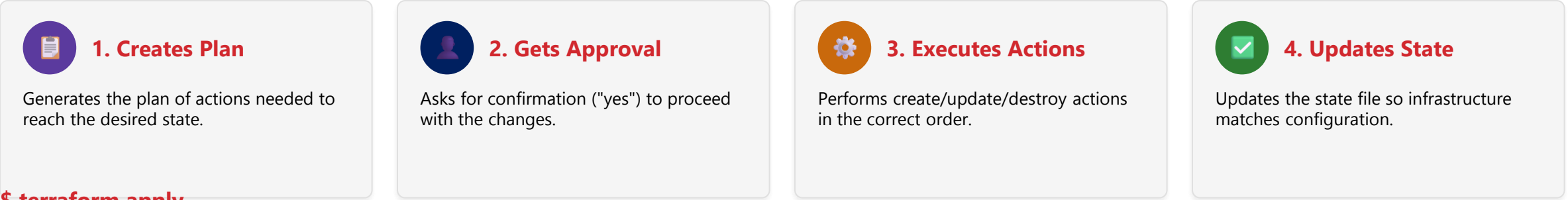
PLAN EXIT CODES

#	Meaning
0	No changes -- infrastructure is up-to-date.
1	Changes present -- plan shows actions to be taken.
2	Error -- plan failed.

KEY TAKEAWAY terraform plan is a read-only operation. It shows what will be added, changed, or destroyed -- always run plan before apply, and always read every action.

TERRAFORM APPLY

The terraform apply command executes the actions proposed in the plan and makes the real changes to your infrastructure.



\$ terraform apply

```
Do you want to perform these actions?  
  Only 'yes' will be accepted to approve.  
  Enter a value: yes  
  
aws_instance.web: Creating... [id=i-0abc123def4567890]  
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

APPLY OPTIONS

Command	Behavior
apply	Interactive -- asks for confirmation.
apply -auto-approve	Skips confirmation (CI/CD only).
apply plan.out	Applies a saved plan file.

KEY TAKEAWAY terraform apply makes REAL changes and requires explicit approval unless auto-approved. Always review the plan before you apply -- plan, then approve, then apply.

TERRAFORM DESTROY

The terraform destroy command safely removes all resources managed by Terraform. Use with caution -- this action is irreversible.



1. Creates Destroy Plan

Generates a plan of resources that will be destroyed.



2. Gets Confirmation

Asks for explicit confirmation before proceeding.



3. Destroys Resources

Removes resources in the correct order to handle dependencies.



4. Updates State

Removes destroyed resources from the state file.

\$ terraform destroy

```
Terraform will perform the following actions:
```

```
# aws_instance.web will be destroyed
- resource "aws_instance" "web" { ... }
```

```
Do you really want to destroy all resources? Enter: yes
Destroy complete! Resources: 3 destroyed.
```

SAFETY BEST PRACTICES

- Always run terraform plan before destroy.
- Use -target carefully -- can cause drift.
- Protect production with approval workflows.
- Keep state in a remote backend with locking.

KEY TAKEAWAY Caution: this action is irreversible. Destroyed resources cannot be recovered by Terraform -- review the plan, confirm intentionally, destroy only what you truly want removed.

TERRAFORM STATE FILES

Terraform state is a record of the infrastructure Terraform manages. It maps your real-world resources to your configuration.

WHAT IS A STATE FILE?

- A JSON file that stores the current state of your infrastructure.
- Tracks every resource Terraform creates and manages.
- Essential for planning, applying, and destroying accurately.

STATE FILE CHARACTERISTICS

- Contains sensitive information (IDs, secrets, ARNs).
- Must be secured and access-controlled.
- Stores resource metadata and dependencies.
- Enables drift detection and recovery.
- Should be versioned and backed up.

WHY STATE FILES MATTER

- **Source of Truth** — Terraform relies on state to know what exists in the real world.
- **Resource Mapping** — maps configuration to actual resources.
- **Change Detection** — enables Terraform to detect drift.
- **Safe Operations** — prevents accidental overwrites and inconsistencies.

COMMON ISSUES

- **State Corruption** — may cause inconsistencies.
- **Manual Changes Outside Terraform** — leads to drift.
- **State Lost or Deleted** — can cause resource recreation risk.
- **Concurrent Changes Without Locking** — causes conflicts.

BEST PRACTICES

- Use remote state for teams.
- Enable state locking.
- Encrypt state at rest.
- Limit access with IAM/RBAC.
- Backup regularly; never manually edit state.

STORAGE (BACKENDS)

Amazon S3

Azure Storage

GCS

Terraform Cloud

KEY TAKEAWAY Treat your Terraform state like a database -- secure it, back it up, and manage it carefully. State is the source of truth for planning, applying, and destroying.

REMOTE STATE CONCEPTS

Remote state stores your Terraform state file in a remote backend instead of locally -- enabling collaboration, locking, security, and scalability.

WHY USE REMOTE STATE?

- **Collaboration** — teams can work together on the same infrastructure.
- **State Locking** — prevents concurrent operations and state corruption.
- **Security** — state is stored securely with encryption and access control.
- **Scalability & Reliability** — remote backends are durable, available, and versioned.

POPULAR REMOTE BACKENDS

Backend	Notes
Amazon S3	Durable, versioned; locking via DynamoDB.
Azure Storage	Blob versioning; locking via blob leases.
Google Cloud Storage	Strong consistency and versioning.
Terraform Cloud	Built-in state, locking, versioning, access mgmt.

STATE FLOW WITH REMOTE BACKEND



Backend configuration example

```
terraform {
  backend "s3" {
    bucket = "my-terraform-state"
    key    = "envs/dev/terraform.tfstate"
    region = "us-east-1"
    dynamodb_table = "terraform-lock"
    encrypt = true
  }
}
```

KEY TAKEAWAY Remote state = Collaboration + Safety (Locking) + Security + Reliability. The state file acts as the single source of truth for all plan/apply/destroy operations.

STATE MANAGEMENT BEST PRACTICES

Terraform state is critical. Following best practices ensures your infrastructure stays safe, consistent, secure, and reliable across teams.



1. Use Remote State

Store state in a remote backend (S3, Azure Blob, GCS, Terraform Cloud) -- never locally in team environments.



2. Enable State Locking

Always enable locking (DynamoDB, blob leases) to prevent concurrent writes.



3. Enable Versioning

Turn on versioning to keep history and enable point-in-time recovery.



4. Encrypt State

Encrypt at rest and in transit (SSE-KMS, TLS).



5. Restrict Access

Apply least-privilege IAM policies -- only those who need it.



6. Do Not Commit State

Never commit .tfstate files to version control.



7. Backup Regularly

Automate backups and periodically test restore procedures.



8-10. Workspaces, No Manual Edits, Monitor

Isolate environments with workspaces; never hand-edit state; monitor access and changes.

KEY TAKEAWAY Golden Rule: treat your Terraform state like production data -- because it is. A corrupted or lost state can lead to outages or unwanted resource recreation.

TRY IT YOURSELF — EXERCISE 2: TERRAFORM CHECKS

Reinforces: fmt/init/validate/plan expectations and a remote-state narrative -- planning the checks before ever touching real cloud credentials.

STEPS (notes/lab45-terraform-checks.md)

Step	What To Do
1 — Command Order	List: terraform fmt, init, validate, plan (or an instructor-safe substitute).
2 — Read the Plan	State explicitly: read create/destroy risk before any apply discussion.
3 — State Narrative	Write 3 bullets on encrypted remote state + locking, without backend credentials.
4 — Evidence Path	Note that sanitized plan snippets belong under notes/screenshots/lab-45/.

Command order to document

```
terraform fmt -check -recursive
terraform init -backend=false
terraform validate
terraform plan -var='environment=dev' -out=tfplan
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 2 before continuing: exercises/exercise-02-terraform-checks.md (workspace: examples/module-45-exercises).

WHAT IS ANSIBLE?

Ansible is an open-source IT automation tool that simplifies configuration management, application deployment, and task automation using agentless, human-readable YAML.

KEY FEATURES

- **Agentless Architecture** — no agents to install or manage on target systems.
- **Simple & Human-Readable** — uses YAML playbooks that are easy to learn and maintain.
- **SSH-Based Communication** — uses SSH (or WinRM for Windows) for secure communication.
- **Idempotent** — ensures systems remain in the desired state.

COMMON USE CASES

Configuration Management

Infra Provisioning

App Deployment

Orchestration

Compliance & Security

SIMPLE PLAYBOOK EXAMPLE

```
- hosts: webservers
  become: yes
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
    - name: Start nginx
      service:
        name: nginx
        state: started
        enabled: yes
```

Ansible pushes configuration and commands to managed nodes over SSH/WinRM and ensures the desired state is achieved.

KEY TAKEAWAY Ansible makes automation simple, powerful, and scalable -- without the need for agents. Reduces manual effort, speeds provisioning, and ensures consistency.

ANSIBLE ARCHITECTURE

Ansible uses an agentless, control node-managed architecture to automate and manage infrastructure at scale.



CONTROL NODE VS MANAGED NODES

Node	Holds / Is
Control Node	Ansible engine, playbooks, inventory, modules, plugins.
Managed Nodes	Linux/Unix, Windows, network devices, cloud instances.

ARCHITECTURE CHARACTERISTICS

- **Agentless** — no software to install on managed nodes.
- **Push-Based** — the control node pushes tasks to nodes.
- **Idempotent** — ensures desired state without side effects.
- **Cross-Platform** — Linux, Windows, network, cloud.

COMMUNICATION

- **Linux/Unix** — SSH (port 22 by default), key-based authentication recommended.
- **Windows** — WinRM (port 5985 HTTP / 5986 HTTPS).

KEY TAKEAWAY Ansible's agentless architecture makes automation simple, secure, and easy to scale -- from a control node to hundreds or thousands of managed nodes.

ANSIBLE INVENTORY FILES

Ansible inventory defines the hosts and groups of hosts on which Ansible runs tasks -- the foundation for targeting systems.

STATIC INVENTORY — INI FORMAT

```
[webservers]
web1 ansible_host=10.0.1.11
web2 ansible_host=10.0.1.12

[dbservers]
db1 ansible_host=10.0.3.11

[all:children]
webservers
dbservers
```

USEFUL COMMANDS

- **Test connectivity** — `ansible all -i inventory.ini -m ping`
- **List hosts in a group** — `ansible webservers -i inventory.ini --list-hosts`

INVENTORY FILE ELEMENTS

- **Hosts** — individual machines or instances (e.g., web1, db1).
- **Groups** — a collection of hosts (e.g., webservers).
- **Variables** — key-value pairs defining connection or configuration settings.
- **Group Hierarchy** — groups can contain other groups ([parent:children]).
- **Special Groups** — all (all hosts), ungrouped (not in any group).

BEST PRACTICES

- Group hosts logically.
- Use variables to avoid repetition.
- Use dynamic inventory for cloud environments.
- Keep sensitive data in Ansible Vault.
- Commit only `inventory.example.yml` -- never real host credentials.

KEY TAKEAWAY Inventory is the backbone of Ansible automation. Organize your hosts, use variables, and choose the right inventory type for scalable, maintainable automation.

ANSIBLE PLAYBOOKS

Playbooks are YAML files that define tasks to run on remote hosts. They describe WHAT you want to achieve -- Ansible handles HOW.

PLAYBOOK STRUCTURE

- **name** — name of the play.
- **hosts** — target hosts or groups.
- **become** — run with privilege escalation.
- **vars** — define variables.
- **tasks** — list of tasks to execute.
- **handlers** — tasks triggered by notifications.

USEFUL COMMANDS

- `ansible-playbook site.yml`
- `ansible-playbook --syntax-check site.yml`
- `ansible-playbook --check site.yml` (dry run)

Example playbook (YAML)

```
---
- name: Install and start Nginx
  hosts: webservers
  become: yes
  vars:
    package_name: nginx
    service_name: nginx
  tasks:
    - name: Install Nginx package
      ansible.builtin.yum:
        name: "{{ package_name }}"
        state: present
    - name: Start and enable service
      ansible.builtin.service:
        name: "{{ service_name }}"
        state: started
        enabled: yes
  handlers:
    - name: Restart Nginx
      ansible.builtin.service:
        name: "{{ service_name }}"
        state: restarted
```

COMMON MODULES

- **package** — install/remove packages.
- **copy** — copy files to remote hosts.
- **template** — copy files from Jinja2 templates.
- **service** — manage services.
- **user** — manage user accounts.

BENEFITS

- Human-readable, version-controllable.
- Reusable and modular.
- Idempotent and consistent.

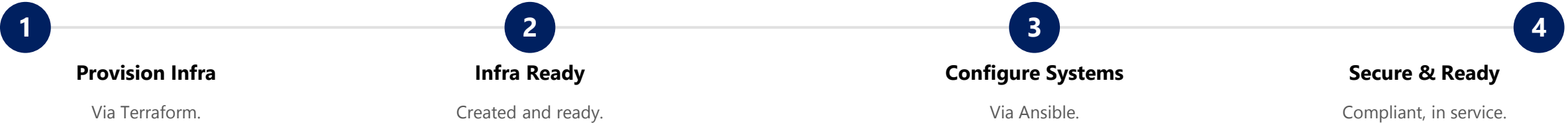
KEY TAKEAWAY Ansible playbooks turn infrastructure tasks into automated, repeatable, and consistent operations. Running the same playbook again produces the same result.

PROVISIONING VS CONFIGURATION MANAGEMENT

Both are essential for automation, but they solve different problems at different stages of the infrastructure lifecycle.

Aspect	Infrastructure Provisioning	Configuration Management
Purpose	Creates and delivers infrastructure resources (VMs, networks, storage).	Configures and maintains software and settings on existing infrastructure.
When It Runs	Done once or infrequently (environment created/changed).	Runs continuously or frequently to keep the desired state.
Common Tools	Terraform, AWS CloudFormation, Azure ARM/Bicep, Pulumi.	Ansible, Puppet, Chef, SaltStack.
Outcome	New infrastructure (instances, VPCs, buckets, DBs).	Configured systems that follow the desired state (apps, configs, policies).
Owned By	Typically platform/DevOps/cloud teams.	Typically operations/SRE/DevOps teams.

HOW THEY WORK TOGETHER



KEY TAKEAWAY Provisioning builds it. Configuration management keeps it running the right way, all the time. Together they deliver reliable, consistent, repeatable environments.

AUTOMATING CONFIGURATION TASKS

Automation helps you configure systems consistently, reduce manual effort, and ensure they remain in the desired state -- every time.



COMMON CONFIGURATION TASKS

- User & group management.
- Package management (install/update/remove).
- Service management (start/stop/enable).
- File & directory management, permissions.
- Security hardening (firewall, SSH, sudoers).
- Application configuration and templates.

Example playbook snippet

```
- name: Configure Web Servers
  hosts: webservers
  become: yes
  tasks:
    - name: Install Nginx
      ansible.builtin.apt:
        name: nginx
        state: present
    - name: Ensure Nginx running
      ansible.builtin.service:
        name: nginx
        state: started
        enabled: yes
      notify: Restart Nginx
  handlers:
    - name: Restart Nginx
      ansible.builtin.service:
        name: nginx
        state: restarted
```

BENEFITS YOU GET

- Consistent configurations across all environments.
- Faster deployments and updates.
- Reduced manual errors and troubleshooting time.
- Easy rollback and version control with Git.
- Auditability and compliance enforcement.

KEY TAKEAWAY Ansible is idempotent -- running the same task multiple times produces the same result without changing the system if it is already in the desired state.

IDEMPOTENT OPERATIONS

Idempotency ensures that running the same operation multiple times produces the same result without unintended side effects.

IDEMPOTENCY IN ACTION — ENSURE NGINX IS INSTALLED

1st Run	2nd Run	3rd Run	Result
changed (installed now)	ok (already present)	ok (already present)	Same desired state every time - - safe to run repeatedly.

HOW ANSIBLE ACHIEVES IDEMPOTENCY

- **State Awareness** — modules check current state before making changes.
- **Declarative Modules** — describe desired state (state: present), not steps.
- **Smart Defaults** — modules return changed only when a change is made.
- **Accurate Reporting** — modules report facts about what changed (or not).

IDEMPOTENT VS NON-IDEMPOTENT

Good (module, declarative) vs bad (raw command)

```
# Idempotent -- checks state before acting
- name: Ensure Nginx is installed
  ansible.builtin.apt:
    name: nginx
    state: present

# Non-idempotent -- always re-runs, no state check
- name: Install Nginx (BAD)
  ansible.builtin.command: apt install -y nginx
```

BEST PRACTICES FOR IDEMPOTENCY

- Use Ansible built-in modules instead of raw shell/command.
- Use state parameters (present, absent, started).
- Use templates, not copy, for config files.
- Test playbooks by running them multiple times.

KEY TAKEAWAY Idempotency is the foundation of reliable automation, self-healing systems, and DevOps at scale. Idempotent operations are safe to run again and again.

TRY IT YOURSELF — EXERCISE 3: ANSIBLE IDEMPOTENCE

Reinforces: naming idempotent modules/handlers for a CRM host sketch and proving a second run reports zero changes.

STEPS (notes/lab45-ansible-idempotence.md)

Step	What To Do
1 — Modules	Name the modules/handlers you expect: package, service, copy/template, handler restart.
2 — Second-Run Rule	State: second run should be changed=0 when authorized; prove with lint/syntax first.
3 — Ownership/Modes	Note that file ownership and modes matter for application config files.
4 — Inventory Hygiene	Commit only inventory.example.yml -- never real host credentials.

Idempotent CRM host sketch (from the lab's worked example)

```
- name: Create CRM service account
  ansible.builtin.user: { name: crm, system: true, shell: /usr/sbin/nologin }
- name: Install runtime configuration
  ansible.builtin.template: { src: crm.env.j2, dest: /etc/crm/crm.env, mode: "0640" }
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 3 before continuing: exercises/exercise-03-ansible-idempotence.md (workspace: examples/module-45-exercises).

TERRAFORM AND ANSIBLE TOGETHER

Terraform provisions infrastructure. Ansible configures it. Together they deliver end-to-end automation for consistent, reliable, repeatable environments.



DYNAMIC INVENTORY WITH TERRAFORM OUTPUT

```
terraform output instance_ips
-> ["34.201.10.21", "34.201.55.10"]
Generated inventory [webserver] section uses those IPs directly.
```

BEST PRACTICES FOR USING TOGETHER

- Keep responsibilities separate and clear.
- Generate inventory automatically from Terraform outputs.
- Run Ansible only after infrastructure is successfully created.
- Design for idempotency in both Terraform and Ansible.

TERRAFORM (PROVISIONING)

- Cloud resources (VMs, VPCs, subnets, security groups).
- Idempotent infrastructure lifecycle.
- Focus: Infrastructure as Code.

ANSIBLE (CONFIGURATION)

- Installs packages, configures files, users, services.
- Deploys applications, enforces policy.
- Focus: Configuration as Code.

COMMON USE CASES

Spin up + configure servers

CI/CD infra + app deploy

Multi-tier app stacks

Ephemeral test environments

KEY TAKEAWAY Terraform builds it. Ansible configures it. Together they automate the full lifecycle -- faster, safer, and more consistently than either tool alone.

INFRASTRUCTURE AUTOMATION BEST PRACTICES

Good automation is not just about tools -- it's about the right practices, processes, and people working together.

1 Version Control Everything

Store all code (Terraform, Ansible, scripts, configs) in Git with branching and pull requests.

2 Modularize and Reuse

Break configurations into small, reusable modules or roles to reduce duplication.

3 Use Variables & Parameters

Avoid hardcoding values -- use variables, tfvars, and inventory/group vars.

4 Follow Least Privilege

Grant only the minimum permissions needed for users, service accounts, and tools.

5 Validate and Test Early

Use linters, validate, and plan/check mode before applying changes.

6 Automate Testing & CI/CD

Run automated tests and pipelines for plan, test, and deployment.

7 Keep State Secure

Store Terraform state remotely with encryption and locking -- protect sensitive data.

8 Observability & Docs

Enable logging/metrics for automation runs; document decisions and runbooks.

KEY TAKEAWAY Golden Rule: automate for consistency, validate for safety, monitor for reliability, improve for the future. Common anti-pattern: manual changes in production.

TRY IT YOURSELF — EXERCISES 4-6: AI REVIEW DISCIPLINE

Capstone: constrain an AI prompt, outline the AI review record, and analyze cost/exposure risk -- the final gate before Lab 45 begins.

STEPS (notes/lab45-ai-*.md and notes/lab45-cost-exposure-quiz.md)

Exercise	What To Do
4 — AI Prompt TODOs	Fill a bounded prompt template (goal, environment, must-include, must-forbid, assumptions) and write one AI suggestion you would reject.
5 — AI Review Record	Outline docs/ai-iac-review.md fields: prompt, generated excerpt, human corrections, validation evidence, rejected item, residual risks, approval.
6 — Cost/Exposure Quiz	For public DB, open security group, oversized VM, and missing tags -- write why each fails the contract.

AI prompt TODO template to fill

```
Goal: _____ Environment: _____ Must include: _____
Must forbid: secrets, public DB, _____ Assumptions: _____
Output files: infra/terraform/*.tf, infra/ansible/site.yml
```

KEY TAKEAWAY TRY IT NOW — Complete Exercises 4-6 before continuing: exercises/exercise-04/05/06-*.md (workspace: examples/module-45-exercises). Final gate before Lab 45.

LAB OVERVIEW — IaC WITH AI ASSISTANCE

Use AI to draft Terraform + Ansible sketches for Northstar CRM non-production environments, then validate, correct, and document every decision.

BUSINESS SCENARIO

- Leadership freezes an IaC rule: AI may accelerate scaffolding, but humans remain accountable for exposure, cost, state security, and idempotence.
- Syntactically valid Terraform that opens a public database still fails the lab; planned apply without reading the plan fails the lab.
- You draft non-production crm-dev/crm-test sketches so later CI/CD promotions land on predictable namespaces and hosts.

LEARNING OBJECTIVES

- Write bounded IaC prompts stating assumptions and forbidding secrets/public DBs.
- Review generated Terraform and Ansible critically -- not just "it linted".
- Model variables, validation, providers, and outputs safely.
- Create idempotent Ansible tasks with modules, handlers, and ownership/modes.
- Run format, validate, lint, and plan checks and interpret create/destroy risk.
- Document AI assumptions, corrections, residual risks, and approval status.

WHAT YOU BUILD

- lab45-crm under ~/java-bootcamp/examples/: infra/terraform/*.tf (versions, providers, variables, main, outputs), terraform.tfvars.example, infra/ansible/site.yml with templates/, inventory.example.yml, and docs/ai-iac-review.md with entry lab45-001 -- prompt, corrections, validation evidence, and at least one rejected or hardened AI suggestion.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan) and CUS-1002 (Ravi Singh) are app-only -- never in .tf/Ansible vars as PII. Estimated time: 2-3 hours, Intermediate difficulty.

LAB TASKS AND DELIVERABLES

lab45-crm -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Define the infrastructure contract: env, region, network, runtime, DB posture, tags, cost limits, forbidden exposures.
2	Draft with constrained AI prompts: small modules, explicit assumptions, no secrets/public DB. Archive prompt as lab45-001.
3	Review Terraform structure: separate providers/variables/resources/outputs; pin provider versions.
4	Secure state and inputs: mark sensitive variables, .tfvars.example only, update .gitignore.
5	Validate the plan: fmt, init -backend=false, validate, plan -- read every create/update/destroy action.
6	Draft Ansible configuration: modules over shell, handlers, ownership/modes; inventory.example.yml only.
7	Test idempotence: --syntax-check, ansible-lint, and a second run showing zero changes (or documented substitute).
8	Write the AI review record: prompt, corrections, validation evidence, one rejected/hardened item, approval.

EXPECTED DELIVERABLES

- infra/terraform/*.tf (structured, pinned providers).
- terraform.tfvars.example (safe placeholders only).
- infra/ansible/site.yml + inventory.example.yml.
- docs/ai-iac-review.md with corrections + evidence.
- Plan/lint evidence; no secrets or state files committed.

RUBRIC (100 MARKS)

- Environment/Structure 10 · Core Impl 30 · Integration/Config 15 · Failure Handling 15 · Verification 10 · Security/AI Review 10 · Docs 10

KEY TAKEAWAY Blind apply of AI output is an honor violation. Committed state/secrets must be remediated before scoring. AI used with no review log loses AI/security marks.

MODULE 45 SUMMARY

In this module, we explored how Infrastructure as Code helps provision and manage infrastructure consistently, securely, and at scale -- with AI assistance to accelerate productivity.

KEY CONCEPTS COVERED

**Infrastructure as Code**
Define infrastructure in code for versioning, review, reuse, and automation.

**Terraform**
Provision cloud resources declaratively and manage them through state.

**Ansible**
Configure and manage systems to the desired state using playbooks.

**End-to-End Automation**
Combine Terraform and Ansible to automate provisioning, configuration, and validation.

**AI Assistance**
Use AI to draft, explain, troubleshoot, and optimize IaC - with humans reviewing every decision.

**Reliability & Consistency**
Idempotent operations, validation, and best practices ensure consistent outcomes.

MODULE FLOW RECAP



KEY TAKEAWAY Infrastructure as Code, combined with configuration management and AI assistance, enables teams to deliver infrastructure that is secure, consistent, and scalable -- with speed and confidence.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of Infrastructure as Code, Terraform fundamentals, and idempotent automation.

1. What is Infrastructure as Code (IaC)?

- A. Managing infrastructure using manual steps
- B. Managing infrastructure through graphical user interfaces
- C. Defining and managing infrastructure using code**
- D. Storing infrastructure in spreadsheets

3. Which tool is best suited for configuration management and application deployment?

- A. Terraform
- B. Ansible**
- C. Git
- D. AWS CloudFormation

2. Which tool is primarily used for provisioning cloud infrastructure?

- A. Ansible
- B. Docker
- C. Terraform**
- D. Jenkins

4. What does "idempotent" mean in automation?

- A. Running a task once only
- B. Getting a different result each time
- C. Multiple runs produce the same desired state**
- D. Manual approval is required every time

KEY TAKEAWAY IaC defines and manages infrastructure using code; Terraform provisions; Ansible configures; idempotent means multiple runs produce the same desired state.

KNOWLEDGE CHECK (2 OF 2)

Four more questions on Terraform configuration, commands, AWS security groups, and Ansible modules.

5. Which file defines Terraform resources?

- A. inventory.ini
- B. playbook.yml
- C. **main.tf**
- D. hosts.yml

7. What is the purpose of a security group in AWS?

- A. Manage IAM users
- B. **Control inbound and outbound traffic**
- C. Store encrypted backups
- D. Manage billing alerts

6. Which command initializes a Terraform working directory?

- A. terraform plan
- B. **terraform init**
- C. terraform apply
- D. terraform destroy

8. Which Ansible module would you use to install a package on a Linux server?

- A. copy
- B. service
- C. **yum**
- D. template

KEY TAKEAWAY main.tf defines Terraform resources; terraform init initializes the working directory; a security group controls inbound/outbound traffic; yum installs packages on Linux.

MODULE 45 COMPLETE

Infrastructure as Code with AI Assistance

You can now provision infrastructure with Terraform, configure it with Ansible, and use AI assistance responsibly -- with humans accountable for every decision.